

BSCS - Sem 5

Subject: Web Technologies

Topic: WML & XSL

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Wireless Markup Language (WML)

Introduction

- o Wireless Markup Language (WML) is a specialized markup language based on XML.
- o It was created specifically for mobile devices such as early cell phones, PDAs, and pagers, which had limited processing power, very small screens, and low-bandwidth connections.
- o WML played an important role in enabling the Wireless Application Protocol (WAP), which allowed users to browse simplified web pages on mobile devices.
- o Unlike HTML, WML was lightweight and optimized for constrained environments. It allowed developers to

create mobile-compatible content at a time when smartphones and modern web browsers were not yet available.

History & Evolution of WML

- o WML was introduced in 1998 - 1999 by the WAP Forum.
- o It is adopted by Nokia, Ericsson, Motorola.
- o It was the very first markup language developed for mobile phones, aiming to bring internet-based services like banking, news, and messaging to mobile devices.
- o To ensure efficiency, WML content was often converted into a binary format known as WMLC before being sent to devices, which reduced data transfer size.
- o Over time, as smartphones emerged with advanced browsers capable of handling standard HTML and CSS, the use of WML declined.
- o Despite this decline, WML is considered a foundation for the modern mobile web, as it introduced the idea of adapting web technologies for mobile environments.

Role of WML in Web Services

- o Web services are systems that allow applications to communicate with each other over the internet using standards such as XML, SOAP, and REST.
- o WML served as the presentation layer for delivering web service outputs to mobile devices.
- o For example, a banking web service could use SOAP or

REST to provide account details, and these details would be formatted in WML so they could be displayed on WAP-enabled phones.

- o In this way, WML made web services accessible to users who only had basic mobile devices, bridging the gap between traditional internet applications and mobile computing.

Working of WML

The working of WML can be understood in steps:

- o A mobile user sends a request using a WAP-enabled phone.
- o The request first reaches the WAP Gateway, which acts as a translator between the mobile network and standard web protocols.
- o The WAP Gateway converts the request into HTTP and forwards it to the web server.
- o The web server processes the request, retrieves the required WML file, and sends back the response.
- o Before the response reaches the mobile device, the WAP Gateway compiles the WML content into WMLC (binary format) to minimize data size and improve transmission speed.
- o The compiled response is sent back to the phone, where the WML browser interprets the content and displays it as cards grouped into decks. Each card represents a small screen of information.

WML Environment

The WML environment consists of several important components working together:

- o Mobile Device: The client that sends and receives requests.
- o WML Browser (Micro browser): A lightweight browser within the device that can interpret WML code and display it as cards.
- o WAP Gateway Encoder: Converts WML files into a compact binary format (WMLC) for efficient transmission.
- o WAP Gateway Decoder: Translates server responses back into a form readable by the device.
- o Web Server: Stores WML files and executes server-side scripts to process user requests.
- o Network/Internet: The medium that connects the device, WAP Gateway, and web server together.

These elements form a client-server model optimized for mobile communication in low-resource settings.

How Model Works

- o The WML model follows a client-server architecture with the WAP Gateway acting as an intermediary.
- o When a user sends a request, the gateway translates it into standard internet protocols.
- o The web server processes the request and returns a response in WML format. This response is then compiled into WMLC by the gateway before being sent back to the mobile device.

- o The WML browser on the device displays the information as cards, which are grouped into decks. This model ensured that even devices with minimal processing power could access online content.

WML Program Structure

A WML program has the following structure:

- o Document Prolog: Declares the XML version and DOCTYPE.
- o Deck: A WML file is called a deck. Each deck contains multiple cards.
- o Card: The smallest unit of WML, representing a single screen/page of information.
- o Body: The content inside the card, which can include text, images, links, and forms.

This structure makes WML both modular and lightweight, ensuring compatibility with limited mobile devices.

WML Emulator

- o Since it was not always practical to test applications on real mobile devices, developers used WML emulators. These tools allowed them to test and debug WML applications on a computer.
- o Popular emulators included Openwave SDK, Nokia WAP Toolkit, and WinWAP Emulator.
- o They provided a virtual environment where developers could see how their WML content would appear on actual mobile phones.
- o This helped improve development speed

and ensured compatibility across devices.

Running WML File on Server

To run WML files, developers needed to:

- o Save the file with the ".wml" extension.
- o Ensure the web server supports the MIME type: text/vnd.wap.wml.
- o Upload the file to the server so it could be accessed via a WAP enabled phone or emulator.
- o The WML browser on the device would then interpret the file and display it as cards.

This process highlights the specialized requirements of serving WML content compared to HTML.

WML vs HTML

While WML and HTML are both markup languages, they were designed for very different environments:

- o WML was created for mobile devices with limited screens and bandwidth, using cards and decks instead of pages and frames. It required a WAP Gateway and used a binary format (WMLC) for efficiency.
- o HTML was created for desktop browsers, supporting large screens, multimedia, and direct interpretation by browsers. It was optimized for high-bandwidth connections.
- o In short, WML was a temporary solution for mobile

internet, while HTML evolved and eventually became suitable for both desktop and mobile environments with the advent of responsive web design.

WML Features

- o XML-Based Structure: WML is based on XML, which means its code is well-organized and follows a clear structure that browsers can easily understand. Example: Just like XML uses tags such as <student> or <name>, WML uses tags like <card> and <p>.
- o Deck and Card Concept: A WML file is called a deck, and it contains several cards. Each card represents one screen or page on a mobile device. Example: A deck named menu. WML may include three cards: Home, About, and Contact.
- o Lightweight Design: WML is specially designed for small screens and slow internet connections, so it loads quickly. Example: Pages opened fast even on old Nokia phones.
- o Event Handling: WML can handle user actions such as clicks, selections, or page loading. Example: Auto-refreshing a page after 5 seconds using a timer event.
- o User Interaction: It allows users to interact through forms, buttons, and links. Example: A login form where the user enters a username and password.
- o Navigation Support: Hyperlinks connect cards so users can move between pages easily. Example:
`<a href="#next">Next</a>` moves to the next card.

WML Elements

The root element that

`<wml>` WML document
wraps the entire `<wml> ... </wml>`

`<card>` Represents one mobile screen or page
Defines a button or
`<card id="home" title="Home Page">`

`<p>` Displays a paragraph or normal text
`<p>Welcome to my site</p>`
`<a href="#about">` About Us `</a>`

`<a>` Creates a hyperlink to another card or URL
`<do type="accept" label="Submit">`
`<do>` Submit or Exit
action such as

<code><input></code>	Accepts user input such as text or numbers	<code><input name="user" title="Enter Name"/></code>
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WML Variables

- Purpose: To store temporary data within a WML deck.
- Declaration: `<setvar name="username" value="Laiba"/>`
- Usage: Displayed as `Hello $(username)!`
- Benefit: Helps personalize content and temporarily

store form data.

◦ Example:

❏ `<setvar name="city" value="Lahore"/>`

❏ `<p>Welcome user from $(city)!</p>`

❏ Output: "Welcome user from Lahore!"

WML Images

◦ WML supports only WBMP (Wireless Bitmap) images — black and white only.

◦ These are used for simple icons, logos, or tiny pictures. ◦ Example: `<p> Welcome!</p>`

◦ Note: No color, no animation — just small monochrome images for faster loading.

WML Tasks

There are three main types of tasks in WML:

◦ Display & Navigation Tasks:

❏ Used to show text or create links between cards. ❏ Example: `<p>Welcome</p> <a href="#next">Go to Next Page</a>`

◦ Input & Form Tasks:

❏ Used to collect data from users.

❏ Example: `<input name="user" title="Enter Name"/>` ◦ Action & Command Tasks:

❏ Used to create buttons that perform actions. ❏ Example: `<do type="accept" label="Submit"> <go href="welcome.wml"/> </do>`

WML Tables

- o Tables in WML are very basic and used only for simple layouts.
 - o They do not support colors, borders, or merged cells.
- Example:

```
o <table>
o <tr><td>Name</td><td>Laiba</td></tr>
o <tr><td>City</td><td>Lahore</td></tr>
o </table>
```

WML Events

Events make WML pages interactive.

Example

ontimer After a certain delay refresh or redirect

Auto

Open a detail card

onpick When a user selects a link

onenterforward	When a card is entered	Show a welcome message
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Example:

- o <card id="home" ontimer="#next">
 - o <timer value="3"/>
 - o <p>Loading next page...</p>
 - o </card>
 - o (This automatically redirects to the next card after 3 seconds.)
- ### WML Server-Side Scripts
- o WML can connect with server-side technologies such as PHP, ASP, JSP, or CGI to create dynamic pages.

How It Works:

- o The user submits a WML form.
- o The WAP Gateway sends the request to the web server.
- o A server-side script (like PHP) processes the data.
- o The server sends back a new WML page.
- o The WML browser displays the updated card.

Example Scenario:

A login page where the user enters credentials → PHP checks them → the next WML card shows "Welcome Laiba!"

Common Use Cases:

- o Online banking
 - o News and weather updates
 - o Data-driven mobile services
- ### Advantages and Limitations of WML

Advantages

- o XML-based → structured and easy to parse.

- o Lightweight → loads quickly on low-bandwidth networks.
- o Optimized for small mobile screens.
- o Allows interactivity through forms and events.

Example: A banking site could load a light WML version on old mobile phones.

Limitations

- o Limited design — no colors, images, or multimedia.
- o Works only on WAP-enabled mobile devices.
- o Outdated — replaced by HTML5 and responsive websites.
- o Summary Line:
- o WML played a key role in the early mobile web era, but it was eventually replaced by modern technologies like HTML5 and CSS for today's smartphones.

XSL (Extensible Stylesheet

Language) Introduction

- o XSL describes how the XML document should be displayed.
- o It allows developers to separate data (XML) from presentation (pdf).
- o Core idea: XML carries raw data; XSL tells how to display and transform it.

Components

- o XSLT (XSL Transformations)
 - Used to transform XML documents to other formats (HTML, etc.)
 - Example: Converting product data stored in XML

- into a styled HTML catalogue
- o XPath (XML path language)
 - 🎬 Query language used to navigate XML document
 - 🎬 Helps XSLT Locate nodes /elements.
 - 🎬 Example: “/bookstore / book/” Title selects all book title
- o XSL - FO (XSL formatting objects)
 - 🎬 Defines how XML data should be formatted for output.
 - 🎬 Less common in web devices but used in publishing.
 - 🎬 Example, Turning XML invoice into PDF bills

Why do we need XSL?

- o Separation of data and presentation: XML for structure, XSL for display
- o Multiple outputs from one source; the same XML as HTML, Pdf, etc.
- o Dynamic web content: XML received from web servers can be
- o styled using XSL.
- o Reusability: one XML dataset, many XSL

stylesheets How XSLT transformation works?

The transformation process follows four steps:

1. XML document -> Source data
2. XSLT style sheet -> Rules for how to transform
3. XSLT processor (like browser engine) -> Applies rule
4. Output document -> can be HTML, PDF, text, etc.

Example

XML file (books.xml)

```
<book store>
<book>
<title> web technologies </title>
<author> Noorayy Jam </author>
<price> 1200 </price>
</book>
<book>
<title> Data Structure </title>
<author> Fatima </author>
<price> 800 </price>
</book>
</book store>
```

XSL file (books.xsl)

```
<xsl: stylesheet version = "1.0"
xmlns: xsl = "https: //www.w3.org/1999/XSL/transform">
<xsl: template match = "/">
<html>
```



```
<body>

<h2> book store catalog </h2>

<table border = "1" cell padding = "5">

<tr bgcolor = "#cccccc">

<th> Title </th>

<th> Author </th>

<th> Price </th>

<xsl: for-each select = "book store / book" >
<tr> <td> <xsl: value-of select = "title"/></td>

<td> <xsl: value-of select = "author"/></td>

<td> <xsl: value-of select = "price"/></td>

</xsl for-each></table></body></html>
</xsl: template></xsl: stylesheet>
```

XPath in Action

Examples:

- o / book store /book /title -> Selects all <title> elements
- o //author -> Selects all <authors> nodes anywhere
- o /book store /book [price >1000] -> Selects books with price > 1000
- o /bookstore /book [last ()] -> Selects last book in

List Role of XSL in web browser

- o SOAP web services often change data in XML -> XSLT is used
- o to convert SOAP messages into readable reports
- o REST APIs with XML response -> XSL helps render them on web pages
- o RSS /Atom feeds -> styled using XSL to display on websites
- o Enterprise Applications -> XSL - FO used to generate invoices, bills, and pdf reports.

Advantages of XSL

- o Separation of Content and Presentation: XSL allows data (content) and design (presentation) to be kept separate. This makes it easy to update one without changing the other. For example, you can change how data looks without touching the data itself.
- o Same XML reused for multiple outputs: The same XML file can be used to produce different results. For example, one XML file can be transformed into an HTML web page, a PDF report, or even a Word document using different XSL stylesheets.
- o Reusable and Maintainable: One XSL file can be used many times on different XML data. If you make a change in the XSL file, the updated design will automatically apply everywhere it is used, which saves time and effort.
- o Supports Accessibility & Internationalization: XSL can be used with screen readers, which helps people with disabilities access content. It also supports different languages, making it useful for international websites and applications.

Disadvantages of XSL

- o Complex: XSL is more complicated than basic HTML or CSS. To use it properly, you need to understand XML, XPath, and XSL syntax.
- o Performance Issues: When working with very large XML files, XSL processing can be slow and may affect performance.
- o Hard to Learn: Beginners often find XSL confusing because it has many rules and requires knowledge of related technologies.
- o Limited Browser Support: Not every browser can process XSL properly. Sometimes you must use server-side tools to transform XML before sending it to the browser.
- o Verbose Code: XSL code is often long and detailed. This makes it harder to write and read compared to simpler languages like CSS.
- o Difficult Maintenance: In large projects, XSL files can become very complex. Debugging and maintaining them can be difficult for developers.
- o Not Always Necessary: For simple XML display tasks, it is easier to use CSS or JavaScript instead of XSL, which saves time and effort.

Applications of XSL

- o Styling XML from Web Services: XSL is often used to transform XML data from web services (like SOAP or REST APIs) into human-readable web pages.
- o Converting RSS/Atom Feeds: RSS or Atom feeds are

- written in XML. XSL can transform these feeds into proper web pages so that users can read news or blog posts in a nice format.
- o Report Generation: XSL can take XML data and create professional-looking reports such as invoices, catalogs, or PDF documents, making it useful for businesses.
 - o Publishing Systems: XSL is widely used in publishing to generate multiple output formats from a single XML source, such as converting data into HTML for the web and PDF for print.

